

Demonstrate CI Integration by running jenkins in a docker container

Step 1 - Initialize a new project folder

```
1RV24MC038_hrshreeraksha@asus:~$ mkdir lab5 && cd lab5
```

Step 2 - Create a Dockerfile and write the following commands

```
1RV24MC038_hrshreeraksha@asus:~/lab5$ cat Dockerfile
FROM jenkins/jenkins:lts
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Step 3 - Create a docker compose file and add the following code

```
1RV24MC038_hrshreeraksha@asus:~/lab5$ cat docker-compose.yml
version: '3.8'

service:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step 4 - Build the jenkins image

```
1RV24MC038_hrshreeraksha@asus:~/lab5$ sudo docker build -t jenkins .
[sudo] password for 1RV24MC038_hrshreeraksha:
[+] Building 2.4s (7/7) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 136B                             0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:ls   2.4s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io  0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                   0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:ls@sha256:7b1c378278279c8688efd6168c25a1c2723a6bdf0420beb5ccefabee3c 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io 0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:8c1169ca1aa613505b8de9e42b4a7811f7733e1bca04626ebe357f6b9162a087 0.0s
=> => naming to docker.io/library/jenkins                       0.0s
```

Step 5 - Run the jenkins image as a container

```
1RV24MC038_hrshreeraksha@asus:~/lab5$ sudo docker run -p 8080:8080 -p 50000:50000 jenkins
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-11-21 04:43:14.998+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2025-11-21 04:43:15.964+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-11-21 04:43:16.032+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1dd333ce2ea; jvm 21.0.10-LTS
2025-11-21 04:43:16.515+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: No JSP Support for /, did not find org.eclipse.jetty.eep.jsp.JettyJspServlet
2025-11-21 04:43:16.591+0000 [id=1] INFO o.e.j.s.DefaultSessionIDManager#doStart: Session workerName=node0
2025-11-21 04:43:17.097+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2025-11-21 04:43:17.217+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oej9n.ContextHandler$CoreContextHandler@3c782d8e{Jenkins v2.528.2,/,b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oej9n.ContextHandler$CoreContextHandler$CoreContextHandler@8072e049{STARTED}}
2025-11-21 04:43:17.230+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bca6fac{HTTP/1.1,(http/1.1)}{0.0.0.0:8080}
2025-11-21 04:43:17.243+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@36a8d46{STARTING}[12.0.25,sto=0] @2963ms
2025-11-21 04:43:17.244+0000 [id=40] INFO winstone.Logger#logInternal: Winstone Servlet Engine Running: controlPort=disabled
2025-11-21 04:43:17.397+0000 [id=39] INFO jenkins.model.Jenkins#init: Starting version 2.528.2
2025-11-21 04:43:17.511+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2025-11-21 04:43:17.526+0000 [id=56] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2025-11-21 04:43:18.466+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2025-11-21 04:43:18.493+0000 [id=57] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2025-11-21 04:43:18.494+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2025-11-21 04:43:18.483+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2025-11-21 04:43:18.483+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2025-11-21 04:43:18.484+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2025-11-21 04:43:18.486+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2025-11-21 04:43:18.720+0000 [id=75] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2025-11-21 04:43:19.171+0000 [id=46] INFO jenkins.install.SetupWizard#init:
[LF]
[LF] *****
[LF] *****
[LF]
[LF] Jenkins initial setup is required. An admin user has been created and a password generated.
[LF] Please use the following password to proceed to installation:
[LF]
[LF] b03e89d0debf4ee78af6cbb461ba50ec
[LF]
[LF] This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]
[LF] *****
[LF] *****
[LF]
2025-11-21 04:43:29.736+0000 [id=37] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2025-11-21 04:43:29.752+0000 [id=39] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2025-11-21 04:43:32.386+0000 [id=75] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2025-11-21 04:43:32.386+0000 [id=75] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
2025-11-21 04:47:21.324+0000 [id=41] INFO winstone.Logger#logInternal: JVN is terminating. Shutting down Jetty
2025-11-21 04:47:21.325+0000 [id=41] INFO org.eclipse.jetty.server.Server#doStop: Stopped oejs.Server@36a8d46{STOPPING}[12.0.25,sto=0]
2025-11-21 04:47:21.327+0000 [id=41] INFO o.e.j.server.AbstractConnector#doStop: Stopped ServerConnector@7bca6fac{HTTP/1.1,(http/1.1)}{0.0.0.0:8080}
2025-11-21 04:47:21.330+0000 [id=41] INFO hudson.lifecycle.Lifecycle#onStatusUpdate: Stopping Jenkins
2025-11-21 04:47:21.332+0000 [id=41] INFO jenkins.model.Jenkins$1#onAttained: Started termination
2025-11-21 04:47:21.339+0000 [id=41] INFO jenkins.model.Jenkins$1#onAttained: Completed termination
2025-11-21 04:47:21.340+0000 [id=41] INFO jenkins.model.Jenkins#cleanUpDisconnectComputers: Starting node disconnection
2025-11-21 04:47:21.350+0000 [id=41] INFO jenkins.model.Jenkins#cleanUpShutdownPluginManager: Stopping plugin manager
2025-11-21 04:47:21.354+0000 [id=41] INFO jenkins.model.Jenkins#cleanUpPersistQueue: Persisting build queue
```

Step 6 - Output

